United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2025 Trimester: Fall

CSE 4165/CSE 465 Web Programming

Total Marks: 30 Duration: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Write a PHP program to design a web page that processes a student's CT and exam marks. The page should take input from an HTML form for three CT marks, a midterm mark, and a final exam mark. The program must select the best two CT marks out of the three, calculate their average, and then compute the total marks by adding the CT average with the midterm and final exam marks. Finally, display the status as 'Passed' if the total marks > 54 otherwise display 'Failed'. For better understanding, refer to the demo input and output. **[10]**

Sample Input (html)	**Sample Output (php)**
CT 1: 12 CT 2: 15 CT 3: 14 Midterm Marks: 22 Final Marks: 32 Calculate Total	Best two CT's total: 29 Midterm marks: 22 Final marks: 32 **Total Marks = 83** Status: Passed

2. Write a PHP program for a simple event-planning system. The program should take three inputs from the user: **number of items sold per day**, **number of days the sale runs**, and **target number of items to be sold**. First, calculate the **total items sold** by multiplying the items sold per day with the number of days. Classify the sales performance using the following conditions: **[10]**

- **Excellent:** total items sold 500 or more
- **Good:** total items sold 300 or more but less than 500
- **Average:** total items sold 150 or more but less than 300
- **Poor:** total items sold less than 150

The program should then compare the total items sold with the target number of items. Finally, display the total items sold, the performance category, the target value, and clearly state whether the sales are **above**, **below**, or **exactly at** the target, along with the difference.

Sample Input (taken using html form)			Sample Output (printed using PHP code)		
Items Sold	Number of	Target	Total Items	Performanc	Result
25	12	300	300	Good	Target met exactly (0
40	10	350	400	Excellent	Above target by 50
15	10	200	150	Average	Below target by 50
8	10	120	80	Poor	Below target by 40

3. Create a database named **campus_library** and a table named **book_loans**. The table tracks student borrowing habits and book conditions. **[10]**

LoanID	StudentName	BookTitle	DaysOverdue	PenaltyFee	Status
101	Abdul	Data Structures	0	0.00	Returned
102	Jabbar	Operating Systems	12	24.00	Overdue
103	Barkat	Discrete Math	5	10.00	Overdue
104	Rahim	Linear Algebra	2	4.00	Overdue
105	Karim	Data Structures	15	30.00	Lost
106	Fahim	Operating Systems	0	0.00	Returned

Answer the following queries

1. Show the total number of books for each Status (*Returned*, *Overdue*, *Lost*), but only include statuses that have more than 1 entry in the table.
2. For any student who has a Status of '*Overdue*' and DaysOverdue less than 7, change their Status to '*Grace Period*' and set their PenaltyFee to 0.
3. For students with a PenaltyFee greater than 20.00, increase their fee by 10% as a "*processing charge*," but only if the final fee does not exceed 50.00.
4. Display each BookTitle and the total PenaltyFee collected/owed for that book. Sort the results so that the book generating the most money appears at the top.

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1. Create a JavaScript application where users can track their daily calorie intake and get real-time feedback on their eating habits. **[10]**

Index.html:

Create a number input field to enter calories consumed per meal, a **"Add Calories"** button, and below that a label field to display total calories, daily goal progress, and feedback messages

Script.js:

1. Set a **daily calorie goal** of **2000 calories**.
2. Take the user's calorie input from the HTML file.
3. Inside a function:
   a) Add the entered calories to a running total.
   b) Display the **total calories consumed so far**.
   c) Display feedback messages based on progress:
      - 0–800 → "You're off to a healthy start!"
      - 801–1600 → "Good progress, keep it balanced!"
      - 1601–1999 → "Almost at your limit!"
      - 2000+ → "Goal reached! Stay mindful!"
   d) Keep track of the **number of entries** the user has made
   e) If the user logs calories **more than 10 times** without reaching the goal, display **"Be cautious of frequent snacking!"**

Sample Input (User Calorie Input)	Sample Output (Feedback)
300, 500, 700	Good progress, keep it balanced!
1000, 800, 300	Goal reached! Stay mindful!
12 entries, total 1500	Be cautious of frequent snacking!

2. You are managing a **movie night event** where you need to calculate how many screens to rent and how much money will be wasted due to empty seats. Write a PHP function that takes three integer inputs: the total number of people attending, the seating capacity of one screen, and the ticket price per seat. The function should calculate how many **complete screens** are needed to accommodate all people (you cannot book partial screens), and how many **empty seats** will remain after everyone is seated. Furthermore, each screen has a **fixed rental cost of 25,000 BDT**, and the amount of money wasted on empty seats must be calculated based on the ticket price multiplied by the number of unused seats. The output should display the total number of screens booked, total empty seats, and total wasted money. **[10]**

Sample Input (from HTML fields)			Sample Output (print on PHP file)		
Attendees	**Seat Capacity**	**Ticket Price**	**Total Screens**	**Empty Seats**	**Wasted Money**
150	60	500	3	30	15,000
280	100	400	3	20	8,000
320	120	350	3	40	14,000

3. Create a database named **'sundarban'** and a table named **'sales_data'** with the following structure and values: **[10]**

SaleID	ProductName	CategoryID	CategoryName	Quantity	Revenue
1	Laptop	301	Electronics	5	350000
2	Mouse	301	Electronics	15	45000
3	Chair	302	Furniture	8	64000
4	Desk	302	Furniture	6	72000
5	Bottle	303	Accessories	20	30000
6	Pen	303	Accessories	25	20000

Now, do the following (write full PHP–MySQL code):

1. Display the **total revenue per category** from the database.
2. If a product's revenue is below **40,000 BDT**, update its category to **"Low Performing"**.
3. For all products generating more than **70,000 BDT**, add a **10% bonus revenue** to their total.
4. Display each **product's name** along with its **category name** and a label "**Top Seller**" if its revenue is above the average revenue of its own category. Otherwise, display "**Regular Seller**".

United International **University (UIU)**
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1. **JAVASCRIPT:** Create a JavaScript application where users can test their password strength and get real-time feedback. **[10]**

index.html:
Create a password input field (input type password), a "Check Strength" button, and below that a label field to display strength feedback and scoring results.
script.js:

1. Generate a target strength score of 100 points for a "Very Strong" password.
2. Take the user's password input from the HTML file.
3. Inside a function:
    - Calculate password strength based on these criteria:
        - Length: +10 points for every 2 characters (minimum 6 characters required)
        - Uppercase letters: +15 points if present
        - Lowercase letters: +15 points if present
        - Numbers: +20 points if present
        - Special characters (!@#$%^&*): +25 points if present
    - Display strength levels: "Very Weak" (0-30), "Weak" (31-50), "Medium" (51-70), "Strong" (71-90), "Very Strong" (91+)
    - If password reaches 100+ points, display "Perfect Password!" and show a success message
    - Keep track of how many attempts the user has made
    - If the user makes more than 8 attempts without reaching "Strong" level, display "Need practice!" and suggest password tips

Sample Input	Sample Output
abc	Very Weak
hello123	Weak
Strong@Pass	Strong
Perfect@Pass123	Perfect Password!

2. **PHP:** It's time to organize the annual UIU Tech Fest, and you need to calculate the budget for venue booking. Write a PHP function that takes three integers as input: the number of expected attendees, the cost per person for venue arrangement, and the maximum capacity of available venues. The function should calculate the minimum number of venues needed to accommodate all attendees (you must book complete venues; you cannot book partial venues). The function should also calculate how many empty seats will remain after all attendees are seated. Moreover, if each venue costs 15,000 BDT regardless of occupancy, how much money is wasted on empty seats? **[10]**

Sample Input (from html file text fields)			Sample Output (print on PHP file)		
Attendees	Cost per Person	Venue Capacity	Total Venues	Empty Seats	Wasted Money (BDT)
250	120	80	4	70	8,400

180	150	100	2	20	3,000
300	100	75	4	0	0

3. **PHP & MYSQL:** Create a database named 'uiutech_final' and a table named 'employee_final'. The employee table has the following structure and values: **[10]**

EmployeeID	EmployeeName	DepartmentID	DepartmentName	Salary	PerformanceRating
1	Arif Rahman	201	Software Development	45000	B
2	Marium Khan	201	Software Development	52000	A
3	Sabbir Hossain	202	Quality Assurance	38000	C
4	Samira Begum	203	UI/UX Design	42000	B

Now, do the following (write full code, not just SQL queries):

1. Show the total number of employees who received each performance rating (A, B, C, D) across all departments.
2. If an employee has a salary below 40,000 BDT and their current performance rating is not 'D', change their performance rating to 'C'.
3. If an employee has a salary greater than 50,000 BDT, add a 5,000 BDT bonus to their salary, but only if the resulting salary is less than or equal to 60,000 BDT.
4. For each department, display the department names and the number of employees working in that department, sorted by the number of employees (largest department first).

Sample SQL Query

```
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";
$conn = new mysqli($servername, $username, $password,
$dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);
if ($result->num_rows > 0) {
  while($row = $result->fetch_assoc()) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
$conn->close();
```

```
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";
$conn = new mysqli($servername, $username, $password,
$dbname);
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";
if ($conn->query($sql) === TRUE) {
  echo "Record updated successfully";
} else {
  echo "Error updating record: " . $conn->error;
}
$conn->close();
```

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1. Create a JavaScript game where a user tries to guess a secret number generated by your code. Strategies are as follows: **[10]**

Index.html:

Create a text filed (input type text) to take the users' guess and a submit button, below the submit button there is a label field to show the feedbacks and results.

Script.js:

1. Generate a random secret number between 500 and 5000.
2. Take the user's guess from the input field in the HTML file.
3. Inside a function:
   a) Compare the user's guess to the secret number.
   b) Display "Too high!" or "Too low!" in the feedback area of your web page.
   c) If the guess is correct, display "Correct!" and optionally disable the input field to prevent further guesses.
   d) Keep track of the number of guesses the user has made.
   e) If the user runs out of guesses (after 5 attempts), display "Out of guesses!" and optionally disable the input field.

Sample Input	**Sample Output**	
User Guesses (from html)	**Output/Feedback** (on html file using JS file)	**Result**
50, 25, 35, 38, 37	*(suppose random number:37)* Too high!, Too low!, Too low!, Too high!, Correct!	Correct!
10, 50, 90, 80, 85	*(suppose random number:82)* Too low!, Too low!, Too high!, Too low!, Too high!	Out of guesses!
1, 5, 10, 20, 15	*(suppose random number:15)* Too low!, Too low!, Too low!, Too high!, Correct!	Correct!

2. **PHP:** **[10]**

It's the end of the Web Programming course, and we want to celebrate it by having a pizza party. You need to figure out how many pizzas to order and how many slices will be left over. Write a PHP function that takes three integers as input: the number of students in the class, the number of slices each student wants, and the number of slices each pizza has. The function should calculate the minimum number of pizzas you need to order to ensure every student gets enough pizza (you must order whole pizzas; you cannot order fractions of pizzas). The function should also calculate the number of pizza slices that will be left over after all the students have had their fill. Moreover, if the price of each pizza is 1050 BDT (irrespective of size), how much money was wasted based on the leftover slices?

Sample Input (from html file text fields)			**Sample Output** (print on PHP file)		
Number of Students:	Slices per Student:	Slices per Pizza:	Total Pizzas:	Leftover Slices:	Wasted Money (BDT):
10	3	8	4	2	262.5
7	2	6	3	4	700
12	2	4	6	0	0

3.	**PHP & MYSQL:**	**[10]**

Create a dataset named 'uiuweb_final' and a table named 'student_final'. The student table has the following structure and values:

StudentID	StudentName	CourseID	CourseTitle	Grade	LetterGrade
1	Karim Uddin	101	Web Programming	85	B
2	Rahim Ahmed	101	Web Programming	92	A
3	Jashim Hossain	102	Project Management	78	C
4	Jasica Ahmed	101	Web Programming	65	D
5	Faria Karim	102	Project Management	95	A
6	Niassoh Dihan	103	System Analysis and Design	80	B

Now, do the following:

1. Show the total number of students who received each letter grade (A, B, C, D) across all courses.
2. If a student has a grade below 75 and their current letter grade is not 'D', change their letter grade to 'C'.
3. If a student has a grade greater than 80, add 5 bonus points to their grade, but only if the resulting grade is less than or equal to 90.
4. For each course, display the course titles and the number of students enrolled in that course, sorted by the number of students (most popular first).